(4) (c)

Error function of logistic Regression

$$E(w) = -\sum_{n=1}^{N} \{ t_n \operatorname{Log} a_n + (1-t_n) \operatorname{Log}(1-a_n) \}$$

where $a_n = \sigma(w^T x)$

gradient of $E(w)$

$$\nabla_w F(w) = \sum_{n=1}^{N} (a_n - t_n) x_n = x^T (a-t)$$

$$\nabla_w^2 E(w) = \sum_{n=1}^{N} a_n (1-a_n) x_n \cdot x_n^T$$

$$= x R x^T \quad \text{—} (1)$$

where $R_{n \times n}$ is a diagonal matrix of $N \times N$ elements,

i.e $R_{n \times n} = a_n (1-a_n)$ for nth Row.

from equ^n 1

$$H = x_n R x_n^T$$

Multplying by U^T & V from both sides in above equation, where V is a random vector such that $V \neq 0$.

$$\Rightarrow V^T H V = V^T x_n R x_n^T V$$

$$= V^T x_n R (V^T x_n)^T$$

$$= \| V^T x_n R^{1/2} \|^2$$

$$\Rightarrow V^T H V > 0$$

therefore H is positive definite

~~from using theorem~~

$\rightarrow$ Error function $F(w)$ is a convex function

$\Rightarrow$ There exist a unique solution, i.e unique minima.